

HYPER-CONVERGED INFRASTRUCTURE

Get Ready For The Cloud Era

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Hyper-convergence is the current and potentially future course of a journey that began years ago with server and storage virtualisation to optimise the traditional siloed approach to information technology

Hyper-convergence is basically an emerging technology in which the server, storage and networking equipment are all combined into one software-driven appliance. Hyper-converged infrastructure (HCI) can be managed comfortably through an easy-to-use software interface. Typically, a hyper-converged platform uses standard off-the-shelf servers and storage, and a virtualisation hypervisor that helps abstract and manage these resources conveniently. A simple operational structure makes this infrastructure flexible, easy to scale and manage.

HCI is gaining traction not just in enterprises but also with others who can benefit from heavy computing without complex management, such as professional racers! Training and racing is usually data-intensive. Racers need information like speed, acceleration and torque in real time to understand how they are performing and how they need to proceed. One mistake could cost not just a victory but a life too! Racing speed depends on the speed and reliability of the data processing and analytics that takes place behind the scenes. Last year, Formula One racing team Red Bull Racing replaced its legacy systems with HPE Simplivity hyper-converged infrastructure, achieving 4.5 times faster performance, increased agility and lower TCO through the move.

A few quick facts about HCI

At the outset, hyper-convergence might sound similar to all the virtualisation stuff you have read about in the past. So, let us

first set out some facts about HCI before rounding up the latest updates:

1. Nutanix, a leader in HCI, defines that the technology streamlines deployment, management and scaling of data centre resources by combining x86-based server and storage resources with intelligent software in a turnkey software-defined solution.

2. HCI combines compute, storage and network together as a single appliance. It can be scaled out or expanded using additional nodes. That is, instead of scaling up the traditional way by adding more drives, memory or CPUs to the base system, you scale out by adding more nodes. Groups of such nodes are known as clusters.

3. Each node runs a hypervisor like Nutanix Acropolis, VMware ESXi, Microsoft Hyper-V or Kernel-based Virtual Machine (KVM), and the HCI control features run as a separate virtual machine on every node, forming a fully distributed fabric that can scale resources with addition of new nodes.

4. Since most modern HCI are entirely software-defined, there is no dependency on proprietary hardware.

5. HCI does not need a separate team as it can be managed by any IT professional. This makes it ideal for small and medium enterprises.

6. HCI is different from the traditional server and shared-storage architectures. It also differs from the previous generation of converged infrastructure.

7. HCI is not the same as cloud computing!